

Secop Interface Description Document for

AEO
Frequency Control
Dual Wire Serial interface
Single wire Secop Modbus

Variable Speed Drives for NLV, and SLVE Compressors

1 Document Information

1.1 Revision History

Version	Init.	Date	Description	Accept	Init.
1.00	HEF	2016-12-06	Release of document	N	
1.10	HEF	2017-04-07	Update of registers and serial interface	N	
1.11	HEF	2017-10-10	Update of automatic input detection	N	
1.12	HEF	2022-04-08	Clarification of Modbus interface added	N	
2.0	THM	2023-10-19	Update for LC 2 nd generation	N	
2.1	THM	2025-04-15	Update AEO parameter	N	

In Accept column state type of review: I = Formal Inspection, W = Walkthrough, N = No review

1.2 Contributors

Name	Init.
Allan Slot	AS
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1.3 References

ID	Name	Reference
[1]	Customer Requirement Document	Customer Product Description Document - All Documents
[2]	MODBUS over Serial Line Specification and Implementation Guide V1.02	www.Modbus.org
[3]	MODBUS APPLICATION PROTOCOL SPECIFICATION V1.1b3	www.Modbus.org

1.4 Purpose of document

This document describes the input requirements for the CCD Variable speed drives for NLV and SLVE compressors.

1.5 Definitions and abbreviations

Term	Definition
AEO	Adaptive Energy Optimisation
Modbus	A serial bus communications protocol
CCD	Cool Capacity Drive

2 Automatic input detection

The Secop CCD inverters are equipped with four different inputs for easy integration. Almost any temperature controller can be used to control the speed without need of changing the setup. The inverters have automatic input detection and will automatically select the input which is active:

1. Dual Wire Interface (separated RX and TX lines)
2. Frequency signal
3. AEO (Thermostatic operation)
4. Single Wire Interface w. Modbus (Tool4Cool®)

If more signals are connected, the input with highest priority (1-4) will be used.

Modbus input has lowest priority and can be used for monitoring in combination with the other inputs.

If Tool4Cool® sends an active start command, the Modbus input will change priority to 1 and overrule all other input signals. The Modbus input will then remain selected until Tool4Cool® is closed.

2.1 Usage

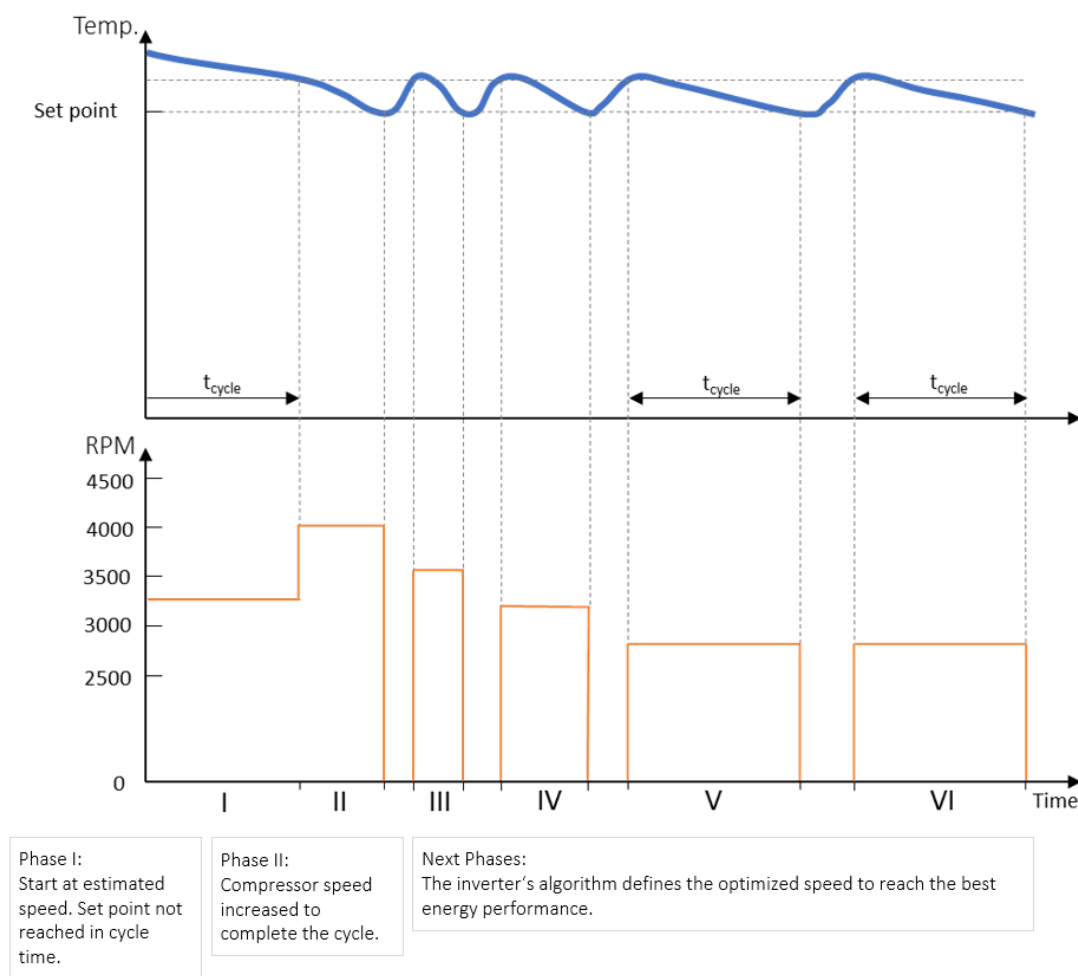
- AEO - For most typical application with a target on/off time, SECOP ECU adjust the speed automatically, customer don't need to change the original thermostat, a wet contact input (1mA?) to the AEO port for on/off.
- Frequency - Customer display (electronic thermostat) adjust the compressor speed as a master controller.
- Dual Wire Interface - Bidirectional communication protocol with separated RX and TX for customer display in mass production with basic parameters, standard display in market and simple to implement.
- Single Wire Secop Modbus interface - Serial communication signal wire in half-duplex mode, major for SECOP T4C with all parameters in developing phase, pay attention on EEPROM registers and timing if used as display.

3 AEO input

AEO – Adaptive Energy Optimization is a kind of semi closed loop control. Instead of working with temperature or pressure, it measures the compressor run time between cut-in and cut-out to estimate the load.

The only parameter to set, is the target run-time. If the run-time is shorter than the target, the speed is reduced for the next cycle. If the run-time is longer than the target, the speed is increased.

- Easy migration from on-off cabinets to variable speed cabinet
- The inverter defines by itself the optimal operation condition
- AEO is an adaptive algorithm, optimizing the compressor speed base on the load and ambient conditions, defrosting periods, etc.
- AEO detects power-faults, defrost cycles, load changes and optimizes both speed and run-time



The following table shows the compressor speed ramp up related to the runtime/cycle time.

% Runtime	% Speed
100	105
110	110
120	120
140	130
160	140
190	180
220	225

3.1 Settings for AEO

Several parameter may be set in Tool4Cool® to optimize the application.

Start Speed:

- The default Start Speed is 4500 RPM.
- A high start speed leads to a fast pull down

Target Runtime:

- The default Target Runtime is 45 minutes.
- A shorter runtime leads to faster adaptation to temperature changes.
- A longer runtime could bring the maximum energy savings.

Number Start Speed Runs:

- The default number is 1 cycle.
- This value defines the number of starts with Start Speed.

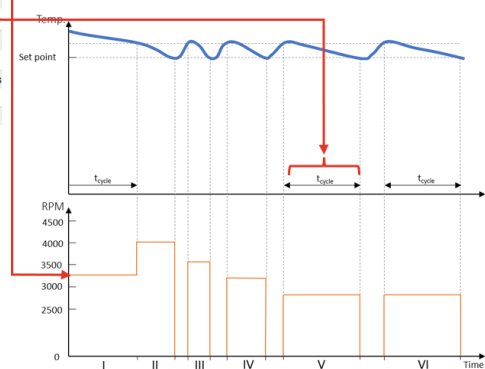
Damping Factor:

- The default value is 90%
- The damping factor lowers the RPM reduction for the next cycle in %.
- 100% → Minimal RPM changes.

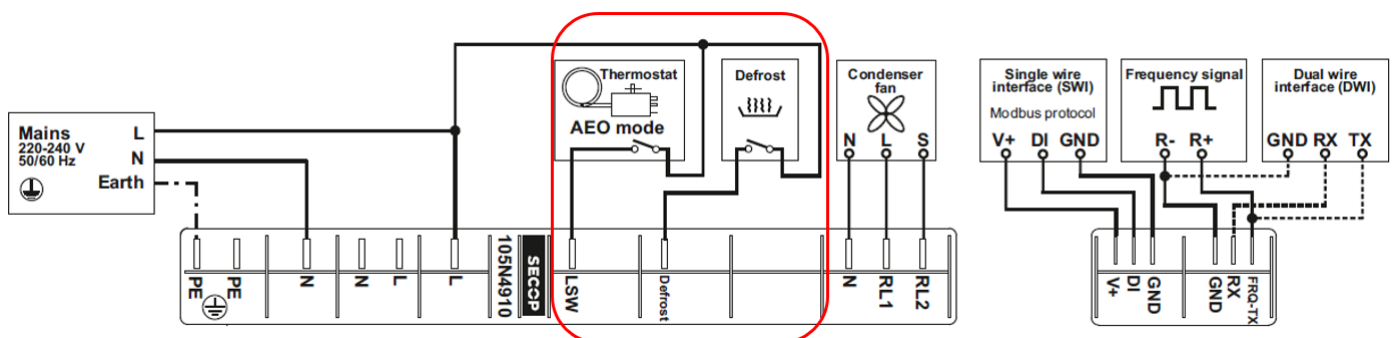
Control Time Base:

- Only for testing purposes. Normal time by default.

Thermostat and AEO	PID	Value
Thermostat		
Thermostat Status	THC01	Open Contact
Thermostat Counter	THC03	0
Reset Thermostat Counter	THC04	RESET THERMOSTAT COUNTER
AEO		
AEO Start Speed	AEO09	3300 rpm
AEO Number Start Speed Runs	AEO09c	1
AEO Damping Factor	AEO02	100 %
AEO Target Runtime	AEO03	45 min
AEO Setup Parameter Transfer	AEO00	TRANSFER AEO SETTINGS
AEO Control Time Base	AEO07	Normal time (1 Minute = 60 Seconds)
AEO Last Runtime	AEO04	00:00:00 d.h:m:s
AEO Last Speed	AEO05a	0 rpm
AEO Next Speed	AEO05b	3300 rpm
Requested speed (based on AEO)	AEO06	0 rpm



3.2 AEO connection



4 Frequency input

The frequency input is used to control the speed of the compressor in closed loop configuration. The frequency control can achieve the same capacity control performance as closed loop control. It is managed by an external thermostat and the frequency signal is used for communication between devices.

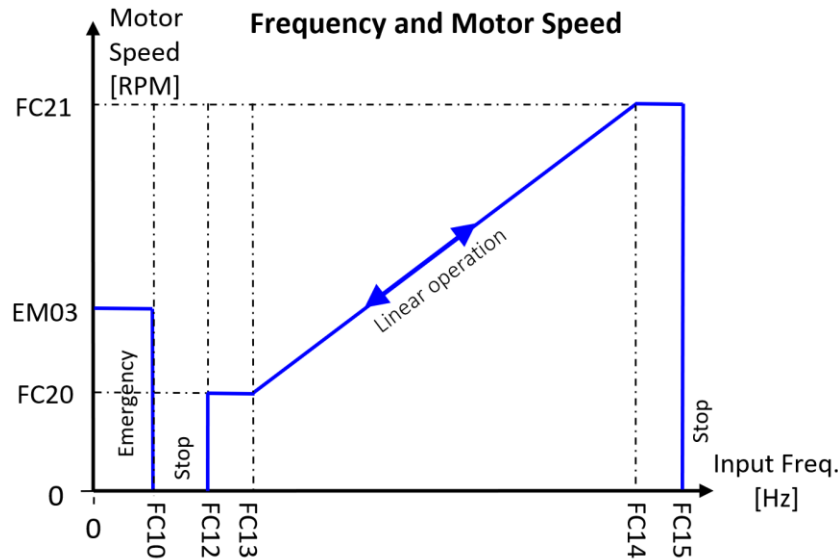


Figure 1: Relationship between the frequency signal and motor speed

The speed of the compressor in the linear area is calculated in the following way:

$$RPM = \left[\left(\frac{FC21 - FC20}{FC14 - FC13} \right) \cdot (F_{in} - FC13) \right] + FC20$$

Example: The Frequency of 66 Hz (FC13) corresponds to 2000 rpm (FC20), 150 Hz (FC14) to 4500 rpm (FC21). This setting leads to a RPM/Frequency rating of 30 rpm/Hz (default setting).

The parameters for the frequency signal and speed are fully programmable and can easily be changed by Tool4Cool®.

Linear operation:

- The speed is changed linear to the frequency between FC13 and FC14
- Certain speeds might be bypassed to avoid resonance and pulsation from pipes and system

Flat operation:

- From FC12 to FC13 and from FC14 to FC15 the compressor operates at fixed speed
- If the compressor is running, it stops if the frequency goes below FC12 or above FC15
- If the compressor is stopped, the compressor starts when the frequency goes above FC12 or below FC15

Compressor stop:

- If the frequency is between FC10 and FC12, or higher than FC15, the compressor will stop
- If no frequency is applied, the compressor will go to emergency mode (Disabled in automatic mode)
- It is recommended to send 25Hz to the inverter when the compressor is stopped

Emergency operation (Disabled in automatic mode):

- If the frequency goes below FC10, the compressor will consider the signal to be missing and go to emergency mode. In this mode, it will run with a fixed speed combined with a duty cycle.
- If Emergency mode is not required: FC10 must be set to 0Hz. In that case, it is recommended to connect the input to GND.

4.1 Settings for frequency interface

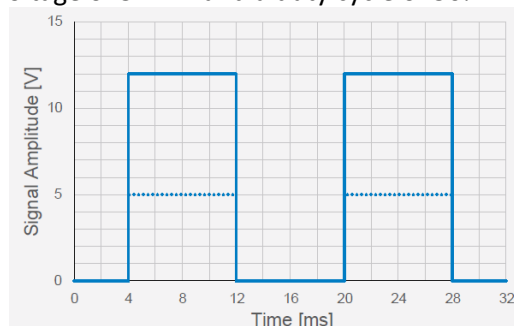
Parameter NLV Function	Code	Min	Max	Default setting	Unit	Accuracy
Frequency Input Signal	FC01				Hz	±1%
Requested Motor Speed	FC02				RPM	
Fallback Frequency	FC10	0	200	10	Hz	
Minimum Frequency	FC12	0	200	50	Hz	
Frequency Ramp Start	FC13	0	200	66	Hz	
Frequency Ramp End	FC14	0	200	150	Hz	
Maximum Frequency	FC15	0	200	200	Hz	
Frequency Ramp End during Defrost	FC16	0	200	150	Hz	
Minimum Speed	FC20	2000	4500	2000	RPM	
Maximum Speed	FC21	2000	4500	4500	RPM	
Maximum Speed during Defrost	FC23	2000	4500	4500	RPM	
Emergency Speed	EM03	2000	4500	2500	RPM	

Parameter SLVE18CN Function	Code	Min	Max	Default setting	Unit	Accuracy
Frequency Input Signal	FC01				Hz	±1%
Requested Motor Speed	FC02				RPM	
Fallback Frequency	FC10	0	200	10	Hz	
Minimum Frequency	FC12	0	200	50	Hz	
Frequency Ramp Start	FC13	0	200	66	Hz	
Frequency Ramp End	FC14	0	200	150	Hz	
Maximum Frequency	FC15	0	200	200	Hz	
Frequency Ramp End during Defrost	FC16	0	200	150	Hz	
Minimum Speed	FC20	2200	4500	2200	RPM	
Maximum Speed	FC21	2200	4500	4500	RPM	
Maximum Speed during Defrost	FC23	2200	4500	4500	RPM	
Emergency Speed	EM03	2200	4500	2500	RPM	

4.2 Signal Requirements

ID	Parameter	Min	Max	Typical	Unit	Accuracy
1	Signal Amplitude (high level)	5	12	5	V _{DC}	±1V
2	Signal Amplitude (low level)	0	1	0	V _{DC}	-1V +0.3V
3	Signal Current	3	8	3	mA	
4	Signal max raise and fall time	0	50	-	µs	
5	Signal missing time before emergency mode	-	30	30	s	±2s
6	Minimum pulse length (F _{IN} ≤ 200Hz)	1.5	-	-	ms	

The frequency signal should have a voltage of 5-12 V and a duty cycle of 50%.

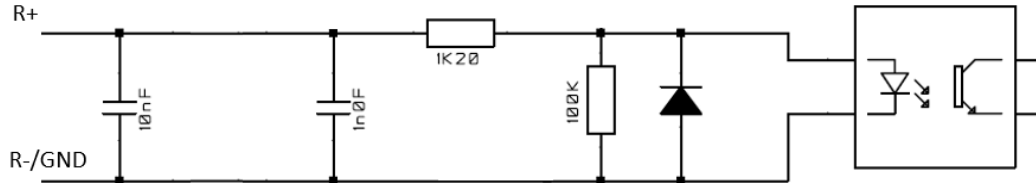


4.3 Schematic of the frequency input

The frequency input shares the input with the DWI/VCC interface.

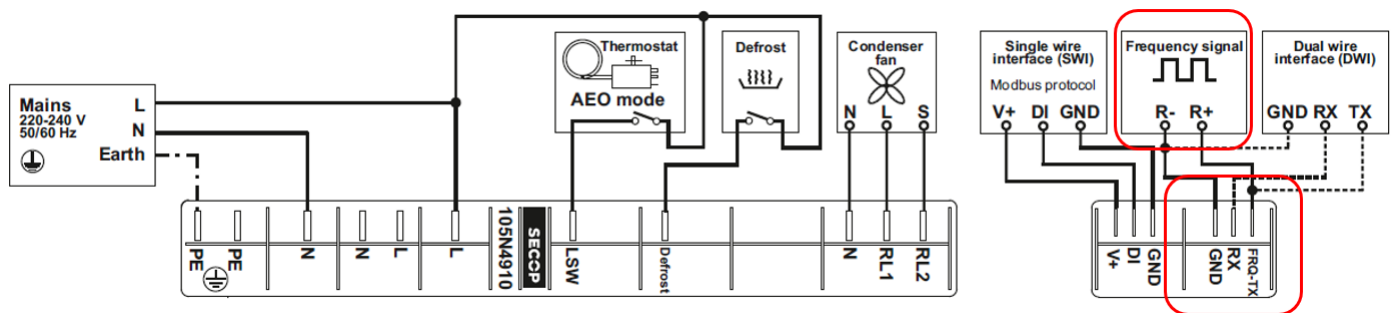
The frequency input is protected for reverse connection. Capacitors and inductors are added to enhance EMC performance.

The following schematic shows an example for the Secop frequency input:



Interface	Name
R+	Frequency Signal
R-	GND

4.4 Frequency Connection



5 Dual Wire Interface (DWI)

The Dual Wire interface is a bidirectional communication protocol with separated RX and TX lines which allows the temperature controller to communicate with the compressor inverter.

The communication interface is shared with the frequency interface. **For proper operation the appliance controller must have a pull-up resistor of approximate 2.2 kohm of the RXD signal.**

Communication Specification	
Baud Rate	600 Baud
Start Bits	1
Data Bits	8
Stop Bits	1
Parity	No
Frame size	5 Bytes
Appliance controller	Master
Compressor controller	Client
Start Bit	1 -> 0 (logic level)
Data Bits	Inverted logic (0V -> "1")
Stop Bit	0 -> 1 (logic level)
Control Mode	Half-duplex

5.1 Command structure

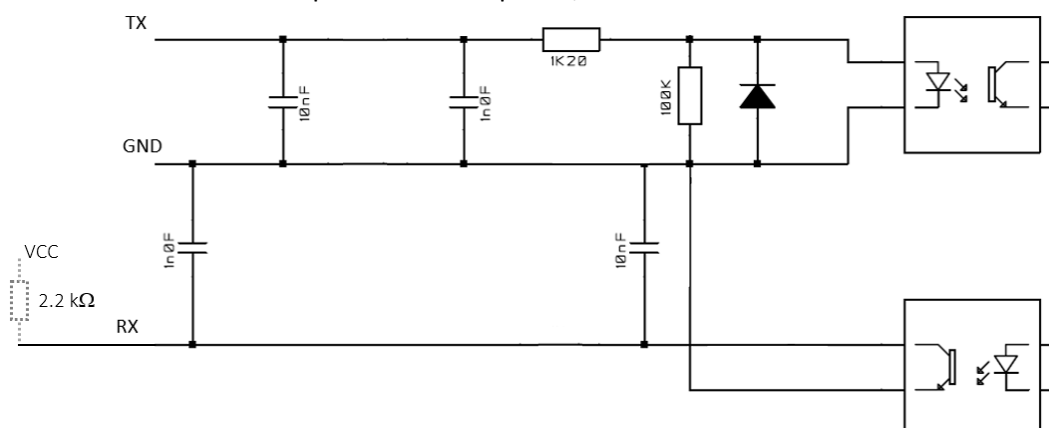
- A protocol consists of 5 data bytes
- The identification byte is used for the synchronization of appliance controller and inverter and initiates the communication
- An identification byte = A5h initiates an interpretation of the protocol by the inverter
- An error-free transmission is acknowledged by the inverter with a defined message
- If a protocol is transmitted with errors, a "Communication Error" is sent as reply

5.2 Schematic of the Dual Wire Interface (DWI)

The the DWI/VCC interface shares the input with the frequency input.

Capacitors and inductors are added to enhance EMC performance.

The following schematic shows an example for the Secop DWI/VCC interface:



Interface	Name
TX	DWI transmit
RX	DWI Receive
GND	DWI GND

5.3 DWI commands

The following table provides an overview over the implemented commands.

Transmit command structure	1st byte	2nd byte	3rd byte	4th byte	5th byte
	Identification	Command	Data low	Data high	Checksum
1: Transmit Set speed	A5h	C3h	1)	2)	3)
2: Read Set speed	A5h	3Ch	80h	39h	66h
3: Read Operation Status	A5h	3Ch	83h	39h	63h
4: Read Actual Speed	A5h	3Ch	86h	39h	60h
5: Read Power	A5h	3Ch	82h	39h	64h
6: Read Temperature	A5h	3Ch	88h	39h	5Eh
7: Read Config Data	A5h	3Bh	00h	01h	1Fh
8: Read Input Voltage	A5h	3Bh	01h	01h	1Eh
9: Read Temperature	A5h	3Bh	02h	01h	1Dh
10: Read Power	A5h	3Bh	03h	01h	1Ch
11: Read Status I	A5h	3Eh	0Eh	00h	0Fh
12: Read Status II	A5h	3Fh	01h	00h	1Bh

Receive command structure	1st byte	2nd byte	3rd byte	4th byte	5th byte
	Identification	Command	Data low	Data high	Checksum
Response to 1	5Ah	83h	4)	5)	3)
Response to 2	5Ah	80h	1)	2)	3)
Response to 3	5Ah	83h	4)	5)	3)
Response to 4	5Ah	86h	7)	8)	3)
Response to 5	5Ah	82h	9)	10)	3)
Response to 6	5Ah	88h	11)	12)	3)
Response to 7	5Ah	00h	00h	00h	A6h
Response to 8	5Ah	01h	13)	14)	3)
Response to 9	5Ah	02h	11)	12)	3)
Response to 10	5Ah	03h	9)	10)	3)
Response to 11	5Ah	3Eh	00h	00h	68h
Response to 12	5Ah	3Fh	00h	00h	67h
Communication Error	5Ah	6)	FFh	FFh	3)

Definition of command bytes:

- 1) Set speed Data low Byte
- 2) Set speed Data high Byte
- 3) Checksum:
 - Sum of Hex-values Bytes 1 - 4 = S14h
 - Checksum = 100h - (S14h AND 0FFh)
- 4) Status Data low Byte
- 5) Status Data high Byte
- 6) Error Code:
 - F0h - Error in Byte 4 (valid for command 2 and 3)
 - F2h - Error Checksum (Byte 5)
 - F4h - Error Command Code (Byte 2)
 - F8h - Error in Byte 3 (valid for command 2 and 3)
- 7) Measured speed Data low Byte
- 8) Measured speed Data high Byte
- 9 Power Consumption low Byte
- 10) Power Consumption high Byte
- 11) InverterTemperature low Byte (°C *10)

12) InverterTemperature high Byte (°C *10)

13) Input Voltage low Byte

14) Input Voltage high Byte

Operation Status

Status	Description
0000h	No failure, compressor running
0020h	Over-temperature (PCB)
0080h	Set Speed data out of range
FF00h	No failure, Compressor stopped
FF01h	Start fail
FF02h	Not used
FF04h	Under speed
FF08h	Wrong rotor position
FF10h	Motor cable fail
FF20h	Overtemperature, compressor stopped
FF40	Serial fail

Status High byte =FFh : Compressor stopped

Status High byte =00h : Compressor running

The register “1: Transmit Set speed” is used to turn on/off the compressor and setting the speed.

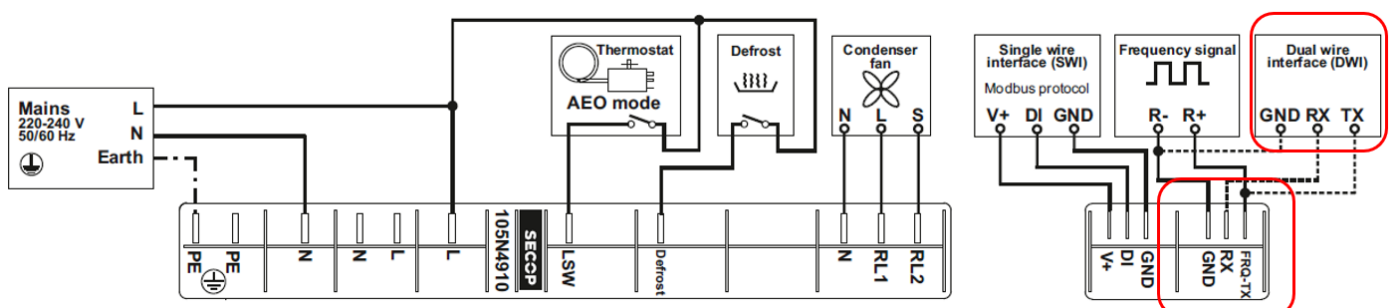
The setting value:

- 0 - 1499 : Compressor stop
- 1500 - 2000 RPM : Out of range
- 2000 - 4500 RPM : Compressor speed set accordingly (2200 RPM min speed for SLVE18CN)
- 4501 – 6000 RPM : Out of range
- > 6000 RPM : Compressor stop

Out of range will result an error in Status, and compressor stop.

To have a succesful starting, the compressor start/stop interval should above 3 minutes when pressure getting balanced, and a soft starting, the first 3 minutes after turn on, set to minimal speed 2000 RPM and ramp up gradually after starting time.

5.4 DWI Connection



6 Single Wire Secop Modbus interface (SWI)

6.1 Hardware interface

The serial communication is implemented as a single wire half-duplex line – transmitting and receiving on the same line.

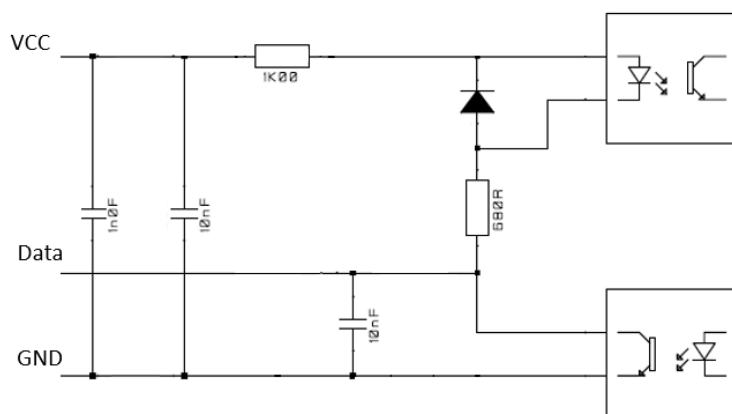
The input port is galvanic isolated from the inverter and must be supplied from the application board by a 5V to 12V DC. The signal level follows the supply voltage. A supply voltage of 12V is recommended.

Up to 3 units can be wired in parallel for multi compressor systems, but it must be ensured that the controller has sufficient drive capability.

Please note:

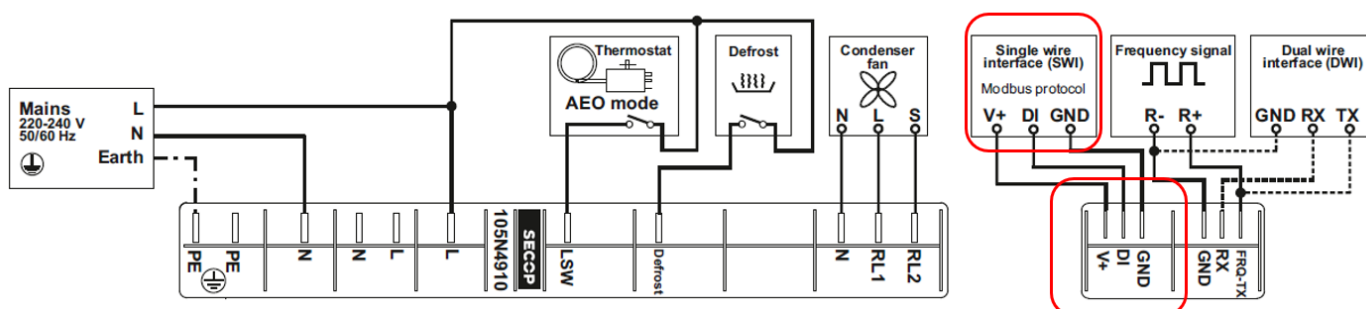
- The interface is not RS485 compatible and cannot handle 2W A/B differential signals, but only logic level signals
- Maximum length of the signal line is 3 meters

Schematic of SWI interface



6.1.1 Electrical data for communication port

Parameter / Limiting values	Min	Max	Typical	Unit
Operating Voltage (VCC)	5	12	5	V DC
Signal Amplitude (high level)	5	12	5	V DC
Signal Amplitude (low level)	0	1	0	V DC
Signal Current	--	3	1	mA
Signal max raise and fall time	0	10	--	μs
Signal missing time before emergency-mode	---	30	10	s



6.2 Communication protocol and Timing

- The MODBUS Serial Line protocol is a Master-Clients protocol. The CCD always operates as a client. A client node will never transmit data without receiving a request from the master node.
- Only one master can be connected to the bus, and up to three CCD's client nodes can be connected to the same serial bus.
- Each CCD must have an individual address which is unique. The CCD's will never communicate with each other.
- The master must always send a message which includes an address – even if only one unit is connected to the bus.
- The client will always return a reply message to the master (unless it is a broad cast message).
- All Modbus transactions consist therefore of two messages – a request from the master and a reply from the CCD.
- No broadcast message is implemented.
- The communication must be refreshed every 10 seconds for safety reasons, If this is not done, the communication is considered as lost and the compressor will stop or go to emergency mode where it will run with a predefined capacity.
- A time-out of at least 200 milliseconds between request and response must be respected in order to give the client time to response.
- All registers are specified according to the Modbus Data Model with registers from 1 to 65535. When addressing, 1 must be subtracted from the given register. (Modbus PDA). For instance, register 35100 must be addressed as 35099.

6.2.1 Settings for the Modbus serial connection

Function	Value
Protocol	Modbus/RTU
Bits per Second (Baud-rate)	9.600
Data bits	8
Stop bits	1
Parity	EVEN
Time-out (ms)	200
Refresh rate (seconds)	10
First network node	1
Last network node	10

6.3 Message Framing

Each message frame must contain address, function code and CRC. Data is optional and depends on message function code.

Message frame

Address	Function code	Data	CRC
8 bits	8 bits	N x 8 bits	16 bits

The Address field contains only the client address.

When the client returns its response, it places its own address in the response address field to let the master know which client is responding.

The function code indicates to the client what kind of action to perform. The function code can be followed by a data field that contains request and response data.

Error checking field is the result of a "Redundancy Checking" calculation that is performed on the message contents. Between each message a silent period of at least 3.5 characters must be obtained.

A message frame must be send as a continuous stream without any silent periods.

6.4 Modbus Function Codes

Supported function codes

Function	Code	Description
Read PNU	0x03	Read one or more PNU/registers. The CCD does not distinguish between “register” and “holding register”. Same function as code 0x04
Read PNU	0x04	Read one or more PNU/holding registers.
Write PNU	0x06	Single PNU/holding register only
Read memory	0x14	Upload of parameter data and log file. Propriety data only
Write memory	0x15	Download of parameter data and log file. Propriety data only
Read device ID	0x2B	Read device identification

6.5 Message Specifications

6.5.1 Read parameter (0x03 / 0x04)

This function is used to read the value of a parameter. Values are always returned as 16-bit integer values and must be scaled according to the parameter definition/scaling in the CCD.

Function code 0x03 or 0x04 retrieves the same data for a specific PNU number.

Request

Function code	1 byte	0x03 / 0x04
PNU Address	2 bytes	0x88B8 - 0xFFFF
Quantity of PNU*	2 bytes	0x0001

Response

Function code	1 byte	0x03
Byte count	1 byte	0x02
PNU data value	2 bytes	0x0000- 0xFFFF

Error

Function code	1 byte	0x83
Error Code	1 byte	1, 2, 3, 4

See Appendix A for error description.

6.5.2 Write single parameter (0x06)

This function is used to write a new setting value to a CCD parameter number. Values must be written as 16-bit integer values and must be scaled according to the parameter definition.

Request

Function code	1 byte	0x06
PNU Address	2 bytes	0x88B8 - 0xFFFF
New PNU value	2 bytes	0x0000 – 0xFFFF

Response

Function code	1 byte	0x06
PNU Address	2 bytes	0x88B8 - 0xFFFF
New PNU value	2 bytes	0x0000 – 0xFFFF

Error

Function code	1 byte	0x83
Error Code	1 byte	1, 2, 3, 4

See Appendix 1 for error description.

6.5.3 Read device ID (0x2B)

This function is used to retrieve the device identification objects: Vendor, Product code and product version.

Request

Function code	1 byte	0x2B
MEI type	1 byte	0x0E
Read device ID code	1 byte	0x01
Device Obj. ID	1 byte	0x00 – 0x02

Response

Function code	1 byte	0x2B
MEI type	1 byte	0x0E
Read device ID code	1 byte	0x01
Conformity level	1 byte	0x01
More follows	1 byte	0x00
Next object ID	1 byte	0x00 - 0x02
Number of objects	1 byte	0x01 - 0x03
Object ID	1 byte	0x00 - 0x02
Object length	1 byte	
Object value	Obj. length	
Object ID		

Object ID for CCD

Object ID	Value
0x00, Vendor	Secop
0x01, Product Code	105Ndddd
0x02, SW Version	yy.zz

Error

Function code	1 byte	0xAB
Error Code	1 byte	1, 2, 3 or 4.

Example

Request:		Response:		
Address	01	Address	01	
Function code	2B	Function Code	2B	
MEI type	0E	MEI	0E	
Device ID code	01	Read Device ID	01	
Device obj ID	00	Conformity level	81	Basic info
CRC	70 77	More follows	00	Only this package
		Next object ID	00	
		Number of objects	03	
		Object ID	00	Vendor
		Object length	05	
		Value (Ascii)	53 65 63 6F 70	SECOP
		Object ID	01	Product code
		Object length	08	
		Value (Ascii)	31 30 35 4E 34 37 30 30	105N4700
		Object ID	02	SW version
		Object length	05	
		Value (Ascii)	30 30 2E 36 32	00.62
		CRC	27 6B	

6.5.4 Modbus Error Codes

Function	Code	Description
01	ILLEGAL FUNCTION	The function code received in the query is not an allowable action
02	ILLEGAL DATA ADDRESS	The data address received in the query is not an allowable address .
03	ILLEGAL DATA VALUE	A value contained in the query data field is not an allowable value
04	SERVER DEVICE FAILURE	An unrecoverable error occurred while the unit was attempting to perform the requested action.

7 Input fault handling

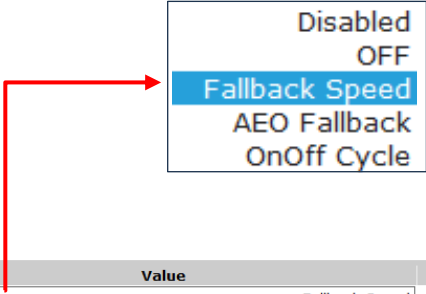
The purpose of this emergency function is to secure cooling in case of loss of connection to the controller.

- The Emergency Mode is active if no frequency or serial communication is detected (if selected)
- The Emergency Mode is not active if Automatic Input Detection is selected

This mode can have 5 different states:

1. The **disabled** mode deactivates the Emergency Mode
 - a. The compressor continues to run at the last set compressor speed
2. **Off**: The compressor will turn off in case of signal lost
3. **Fallback Speed**: The compressor will run at a set emergency speed after a set fallback timeout time.
 - a. Typically high rpm (4500 RPM) for freezer and low rpm (2000 RPM) for cooler
4. **AEO Fallback**: If frequency- or serial input is not detected, the Emergency Mode change to AEO mode
 - a. Used for redundant systems
5. **OnOff Cycle**: If frequency- or serial input is not detected, the Emergency Mode change to a defined On/Off cycle
 - a. For fall backs if minimum RPM is too high

All parameter can be set by Tool4Cool or serial communication.



Input Fault Handling	PID	Value
Emergency Mode Selection	EM01	▼ Disabled OFF Fallback Speed AEO Fallback OnOff Cycle
Fallback Timeout	EM02	1000 sec
Emergency Speed	EM03	3500 rpm

8 Appendix PNU list

Modbus Parameter List for Multipurpose controller									
Modbus register	ID	Parameter Name	Parameter Size/Type	Value Min	Value Max	Value Default	Unit	Read/Write	Comment
Info Registers									
35100	ERR01	Main Error	32bit	0	0xFFFF	-	eNum	R	Main error type 0 = NoMainError 1 = VoltageError 2 = MotorError 4 = TemperatureError 8 = InputError 10 = ElectronicError More errors can be reported simultaneously
35090	ERR02	Sub Error	32bit	0	0xFFFF	-	eNum	R	Sub error type 0 = No error 1- 2047 = Motor errors (bit 1 to 11) 2048 = Inverter too cold (bit 12) 4096 = Inverter too hot (bit 13) 8192 = Heat sink too cold (bit 14) 16384 = Heat sink too hot (bit 15) More errors can be reported simultaneously
35601	ACP01	Power Consumption	16bit	0	2000	-	Watt	R	Power Consumption
35117	ACS01	AC Supply Voltage	16bit	0	300	-	Vac	R	AC Supply Voltage
35119	ACF01	AC Supply Frequency	16bit	0	100	-	Hz	R	AC Supply Frequency
35097	TPI01	Heatsink Temperature	16bit	-20	135	-	°C/100	R	Heatsink Temperature
35099	TPP01	PCB Temperature	16bit	-20	135	-	°C/100	R	PCB Temperature
Input & Modbus parameters									
35901	MDB01	Modbus Address	8bit (EEPROM)	1	247	1	eNum	RW	Modbus node address
35902	MDB02	Bits Per Second	8bit (EEPROM)	9600	9600	9600	Baud	R	Baudrate = 9600
35007	SP01	Speed Control Input	8bit (EEPROM)	4	0	4	eNum	RW	Input Type 0 = Automatic 1 = Frequency 2 = AEO 3 = Modbus 4 = VCC
Emergency Mode parameters									
35023	EM01	Fallback Method	8bit (EEPROM)	4	0	1	eNum	RW	What to do when input signal is missing 0 = Disabled 1 = Stop Compressor 2 = Fallback Speed 3 = AEO Fallback 4 = On- Off-Cycle
35021	EM02	Signal Missing Time Before Fallback	16bit (EEPROM)	10	65535	10	s	RW	Signal missing time before fallback
35046	EM02b	DWI Signal Missing Time Before Fallback	16bit (EEPROM)	10	65535	60	s	RW	DWI Signal missing time before fallback
35022	EM03	Fallback Speed	16bit (EEPROM)	2000	4500	2500	rpm	RW	Speed of compressor when in emergency mode.
35024	EM04	OnOffCycle Minutes ON	16bit (EEPROM)	0	65535	27	min	RW	
35025	EM05	OnOffCycle Minutes OFF	16bit (EEPROM)	0	65535	3	min	RW	
35044	EM06	OnOffCycle Seconds ON	16bit (EEPROM)	0	59	0	sec	RW	
35045	EM07	OnOffCycle Secods OFF	16bit (EEPROM)	0	59	0	sec	RW	
Motor control parameters									
35104		Request Motor Start/Stop	8bit	0	0	1	Boolean	RW	Start/stop the compressor
35098	MS01	Real Motor Speed	16bit	0	4500	-	rpm	R	Read actual motor speed
35033		Set Motor Speed	16bit	0	4500	-	rpm	RW	Requested compressor speed
37100	MS17	MinSpeed	16bit	0	5000	2000	rpm	R	Minimum compressor speed. Model dependent
37101	MS18	MaxSpeed	16bit	0	5000	4500	rpm	R	Maximum compressor speed. Model dependent
AEO									
34816	AEO09	AEO Start Speed	16bit (EEPROM)	2000	5000	4500	rpm	RW	
34822	AEO09c	AEO Number Start Speed Runs	16bit (EEPROM)	1	65535	2	-	RW	
34805	AEO02	AEO Damping Factor	16bit (EEPROM)	0	100	90	%	RW	
34804	AEO03	AEO Target Runtime	16bit (EEPROM)	0	600	45	min	RW	
34801	AEO04	Runtime Seconds	16bit	0	65535	-	s	R	
34802	AEO04	Runtime Minutes	16bit	0	65535	-	min	R	
34819	AEO05a	AEO Last Speed	16bit				rpm	R	
34806	AEO05b	AEO Next Speed	16bit				rpm	R	
34800	AEO06	Requested speed (based on AEO)	16bit				rpm	R	

Defrost									
35263	DF13	Defrost Status	8bit	0	1	0	Boolean	R	Defrost Input
35264	DF14	Defrost Control State	16bit	0	5	0	eNum	RW	0 = No Defrost 1 = Defrost Low Speed 2 = Defrost High Speed 3 = Checking for electrical defrost 4 = Waiting for pulldown 5 = Pulldown Note: Only with AEO
35267	DF15	Defrost Low Speed	16bit (EEPROM)	2000	4500	2000	RPM	RW	
35268	DF16	Defrost Low Speed Time	16bit (EEPROM)	0	60	3	Minutes	RW	
35260	DF10	Defrost High Speed	16bit (EEPROM)	2000	4500	4500	RPM	RW	
35261	DF11	Defrost Pulldown Speed	16bit (EEPROM)	2000	4500	4500	RPM	RW	
35262	DF12	Defrost Off Time Percentage	16bit (EEPROM)	0	25600	0	1/256 %	RW	25600 = 100%
Relays									
36043	RL01	Requested Relay 1 State	16bit	0	1	0	0	RW	Array size on number of relays.
36044	RL02	Requested Relay 2 State	16bit	0	1	0	0	RW	Array size on number of relays.
36053	RL01	Actual Relay 1 State	16bit	0	1	-	0	R	Actual Relay State
36054	RL02	Actual Relay 2 State	16bit	0	1	-	0	R	Actual Relay State
35456	REC04	Relay 1 Lifetime Counters	16bit	0	65535* 65535	-	0	R	Relay Lifetime Activation Counter
35458	REC05	Relay 2 Lifetime Counters	16bit	0	65535* 65535	-	0	R	Relay Lifetime Activation Counter
Condenser Fan									
35740	F11	Condenser Fan Mode	16bit (EEPROM)	0	4	1	eNum	RW	0 = Disabled 1 = Run with Compressor 2 = Always Low Speed 3 = Always High Speed 4 = Low Speed when Compressor stopped
35741	F15	Condenser Fan Speed	16bit (EEPROM)	2000	4500	2500	RPM	RW	
35742	F16	Condenser Fan Speed During Defrost	16bit (EEPROM)	0	2	0	eNum	RW	0 = Off 1 = Low Speed 2 = High Speed
35743	F18	Actual Condenser Fan Speed	16bit	0	2	0	eNum	R	0 = Off 1 = Low Speed 2 = High Speed
35744	F17	Relay Configuration	16bit (EEPROM)	1	3	1	eNum	RW	0 = Relay1HighSpeedOpen 1 = Relay1LowSpeedOpen 2 = Relay2HighSpeedOpen 3 = Relay2LowSpeedOpen
Forbidden Speed Range									
35180	FSR01	1st Forbidden Min Motor Speed	16bit (EEPROM)	0	65535	0	RPM	RW	1st Forbidden Min Motor Speed
35183	FSR02	1st Forbidden Max Motor Speed	16bit (EEPROM)	0	65535	0	RPM	RW	1st Forbidden Max Motor Speed
35181	FSR04	2nd Forbidden Min Motor Speed	16bit (EEPROM)	0	65535	0	RPM	RW	2nd Forbidden Min Motor Speed
35184	FSR05	2nd Forbidden Max Motor Speed	16bit (EEPROM)	0	65535	0	RPM	RW	2nd Forbidden Max Motor Speed
35182	FSR07	3rd Forbidden Min Motor Speed	16bit (EEPROM)	0	65535	0	RPM	RW	3rd Forbidden Min Motor Speed
35185	FSR 08	3rd Forbidden Max Motor Speed	16bit (EEPROM)	0	65535	0	RPM	RW	3rd Forbidden Max Motor Speed
PCB temperature counter									
35350	STH01	PCB Temperature Range1	16bit	0	4x65535	-	Hour	R	Accumulated time @ PCB temp. lower than 0°C
35352	STH02	PCB Temperature Range2	16bit	0	4x65535	-	Hour	R	Accumulated time @ PCB temp. 0°C - 25°C
35354	STH03	PCB Temperature Range3	16bit	0	4x65535	-	Hour	R	Accumulated time @ PCB temp. 25°C - 35°C
35356	STH04	PCB Temperature Range4	16bit	0	4x65535	-	Hour	R	Accumulated time @ PCB temp. 35°C - 45°C
35358	STH05	PCB Temperature Range5	16bit	0	4x65535	-	Hour	R	Accumulated time @ PCB temp. 45°C - 55°C
35360	STH06	PCB Temperature Range6	16bit	0	4x65535	-	Hour	R	Accumulated time @ PCB temp. 55°C - 65°C
35362	STH07	PCB Temperature Range7	16bit	0	4x65535	-	Hour	R	Accumulated time @ PCB temp. 65°C - 75°C
35364	STH08	PCB Temperature Range8	16bit	0	4x65535	-	Hour	R	Accumulated time @ PCB temp. 75°C - 85°C
35366	STH09	PCB Temperature Range9	16bit	0	4x65535	-	Hour	R	Accumulated time @ PCB temp. higher than 85°C
Heatsink temperature counter									
35368	STM10	Heatsink Temperature Range1	16bit	0	4x65535	-	Hour	R	Accumulated time @ Heatsink temp. lower than 0°C
35370	STH11	Heatsink Temperature Range2	16bit	0	4x65535	-	Hour	R	Accumulated time @ Heatsink temp. 0°C - 25°C
35372	STH12	Heatsink Temperature Range3	16bit	0	4x65535	-	Hour	R	Accumulated time @ Heatsink temp. 25°C - 35°C
35374	STH13	Heatsink Temperature Range4	16bit	0	4x65535	-	Hour	R	Accumulated time @ Heatsink temp. 35°C - 45°C
35376	STH14	Heatsink Temperature Range5	16bit	0	4x65535	-	Hour	R	Accumulated time @ Heatsink temp. 45°C - 55°C
35378	STH15	Heatsink Temperature Range6	16bit	0	4x65535	-	Hour	R	Accumulated time @ Heatsink temp. 55°C - 65°C
35380	STH16	Heatsink Temperature Range7	16bit	0	4x65535	-	Hour	R	Accumulated time @ Heatsink temp. 65°C - 75°C
35382	STH17	Heatsink Temperature Range8	16bit	0	4x65535	-	Hour	R	Accumulated time @ Heatsink temp. 75°C - 85°C
35384	STH18	Heatsink Temperature Range9	16bit	0	4x65535	-	Hour	R	Accumulated time @ Heatsink temp. higher than 85°C

Life Time General Reset Counters									
35161	POL01	Low Power Reset Counter	16bit	0	65535	-	eNum	R	
35162	POL02a	Windowed Watchdog Reset Counter	16bit	0	65535	-	eNum	R	
35163	POL02b	Independent Watchdog Reset Counter	16bit	0	65535	-	eNum	R	
35164	POL03	Software Reset Counter	16bit	0	65535	-	eNum	R	
35165	POL04	Power ON Reset Counter	16bit	0	65535	-	eNum	R	
Life Time General Error Counters									
35167	POL06	Voltage Error Counter	16bit	0	65535	-	eNum	R	
35168	POL07	Motor Error Counter	16bit	0	65535	-	eNum	R	
35169	POL08	Temperature Error Counter	16bit	0	65535	-	eNum	R	
35170	POL09	Input Error Counter	16bit	0	65535	-	eNum	R	
Life Time General Heatsink Min/Max Temperature									
35172	BL01	Heatsink Lowest Measured Value	16bit	-327,68	327,67	-	1/100 °C	R	
35173	BL02	Heatsink Highest Measured Value	16bit	-327,68	327,67	-	1/100 °C	R	
Life Time General PCB Min/Max Temperature									
35175	BL04	PCB Lowest Measured Value	16bit	-327,68	327,67	-	1/100 °C	R	
35176	BL05	PCB Highest Measured Value	16bit	-327,68	327,67	-	1/100 °C	R	
Life Time Counter Speed									
35050	WH01	Speed Range1	16bit	0	4x65535	-	Hour	R	1500-2000 rpm Run Time counter during lifetime
35052	WH02	Speed Range2	16bit	0	4x65535	-	Hour	R	2001-2500 rpm Run Time counter during lifetime
35054	WH03	Speed Range3	16bit	0	4x65535	-	Hour	R	2501-3000 rpm Run Time counter during lifetime
35056	WH04	Speed Range4	16bit	0	4x65535	-	Hour	R	3001-3500 rpm Run Time counter during lifetime
35058	WH05	Speed Range5	16bit	0	4x65535	-	Hour	R	3501-4000 rpm Run Time counter during lifetime
35060	WH06	Speed Range6	16bit	0	4x65535	-	Hour	R	4001-4500 rpm Run Time counter during lifetime
35062	WH07	Speed Range7	16bit	0	4x65535	-	Hour	R	4501-5000 rpm Run Time counter during lifetime
35066	WH08	System On Run Time	16bit	0	4x65535	-	Hour	R	System On Run Time counter during lifetime
35064	WH09	Total Motor Run Time	16bit	0	4x65535	-	Hour	R	Total Motor Run Time counter during lifetime